

Use this caption in your report/PPT:

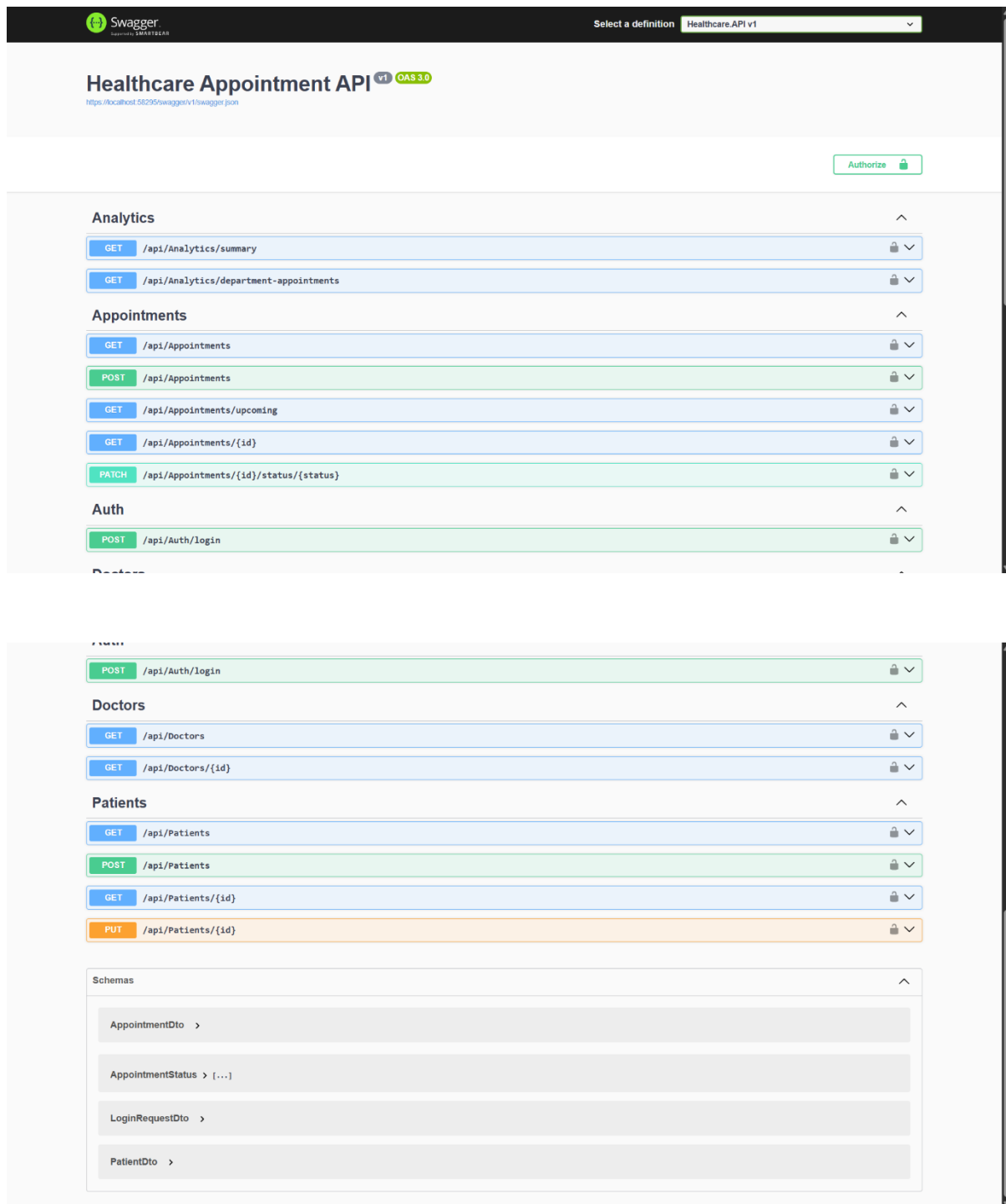


Figure: Healthcare Web API running successfully on localhost.

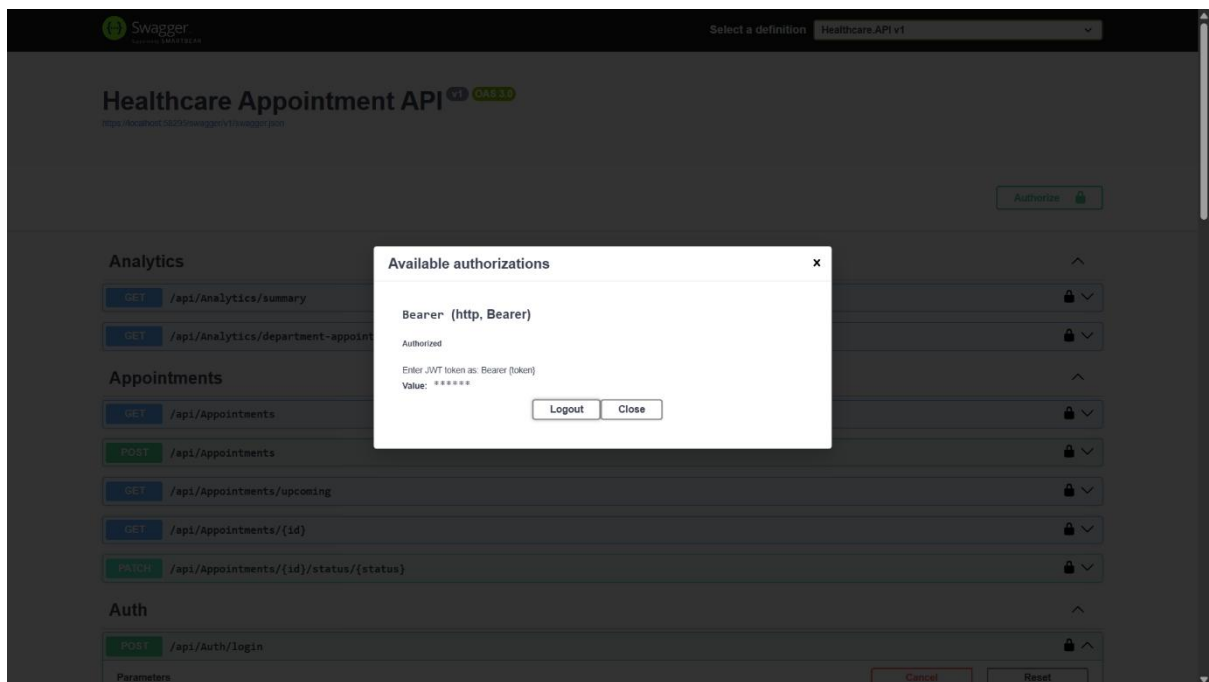
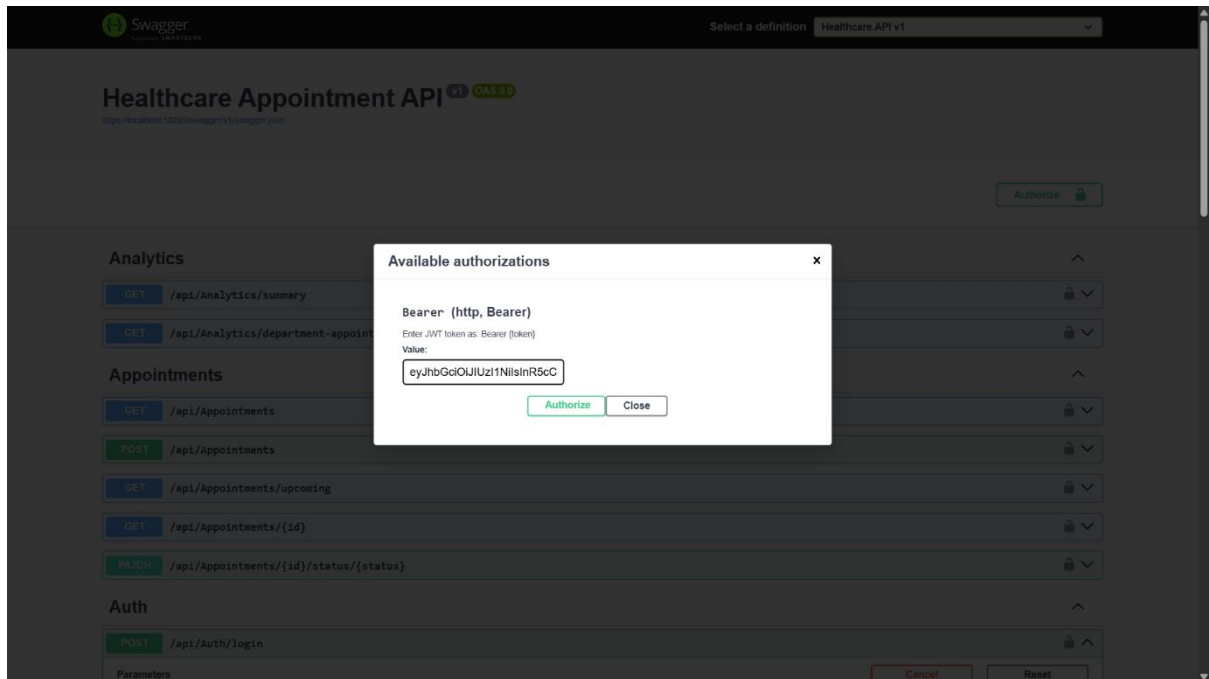


Figure: Swagger authorized using JWT Bearer token.

POST

/api/Auth/login

Doctors

GET

/api/Doctors

GET

/api/Doctors/{id}

Patients

GET

/api/Patients

Parameters

Cancel

No parameters

Execute

Responses

Code	Description	Links
200	OK	No links

POST

/api/Patients

GET

/api/Patients/{id}

PUT

/api/Patients/{id}

Schemas

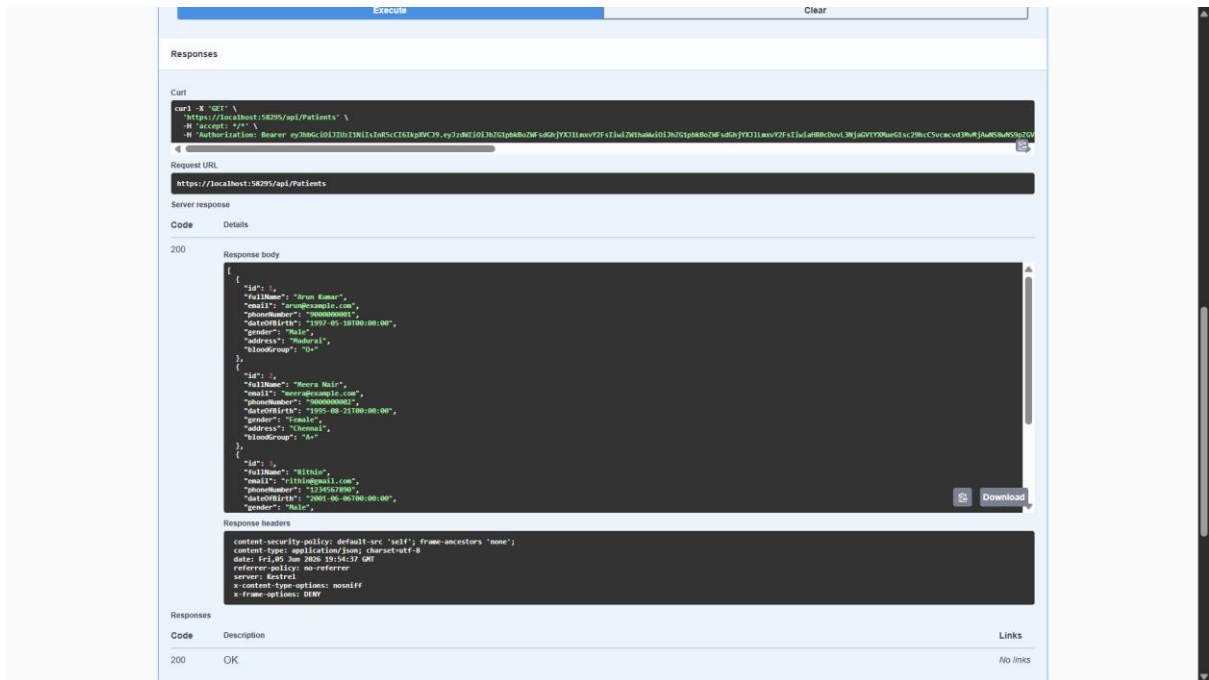


Figure: Patients API returning patient records from SQL Server.

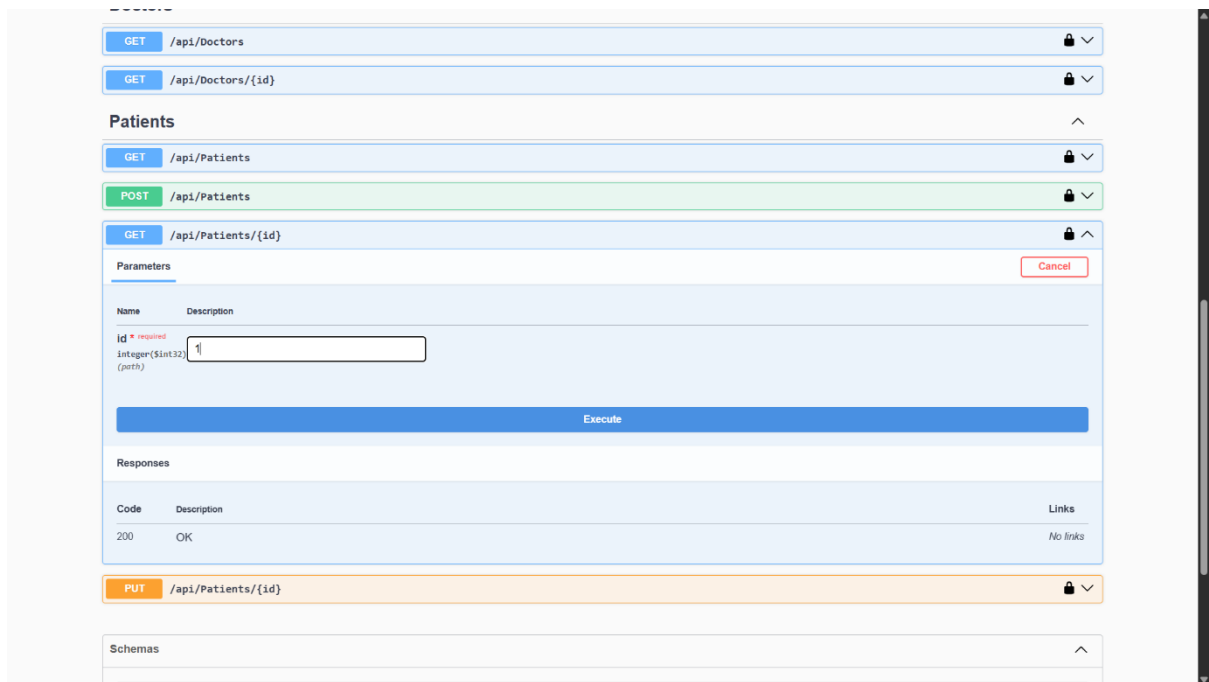


Figure: Patient details retrieved by Patient ID.

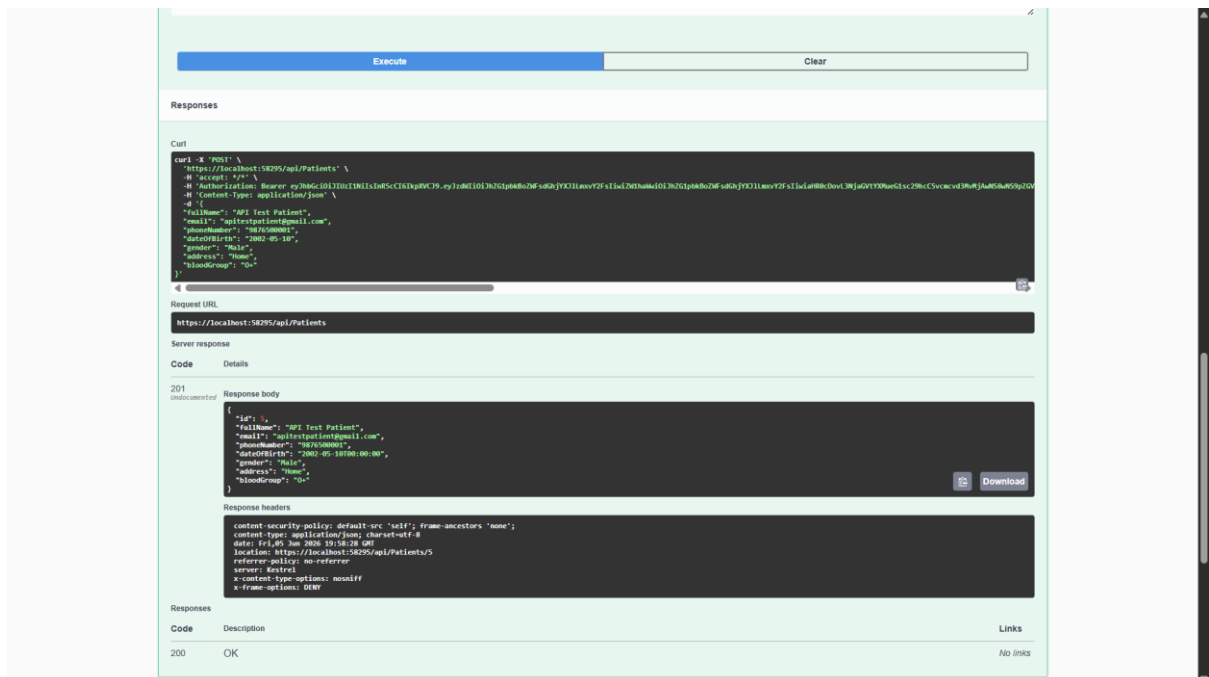
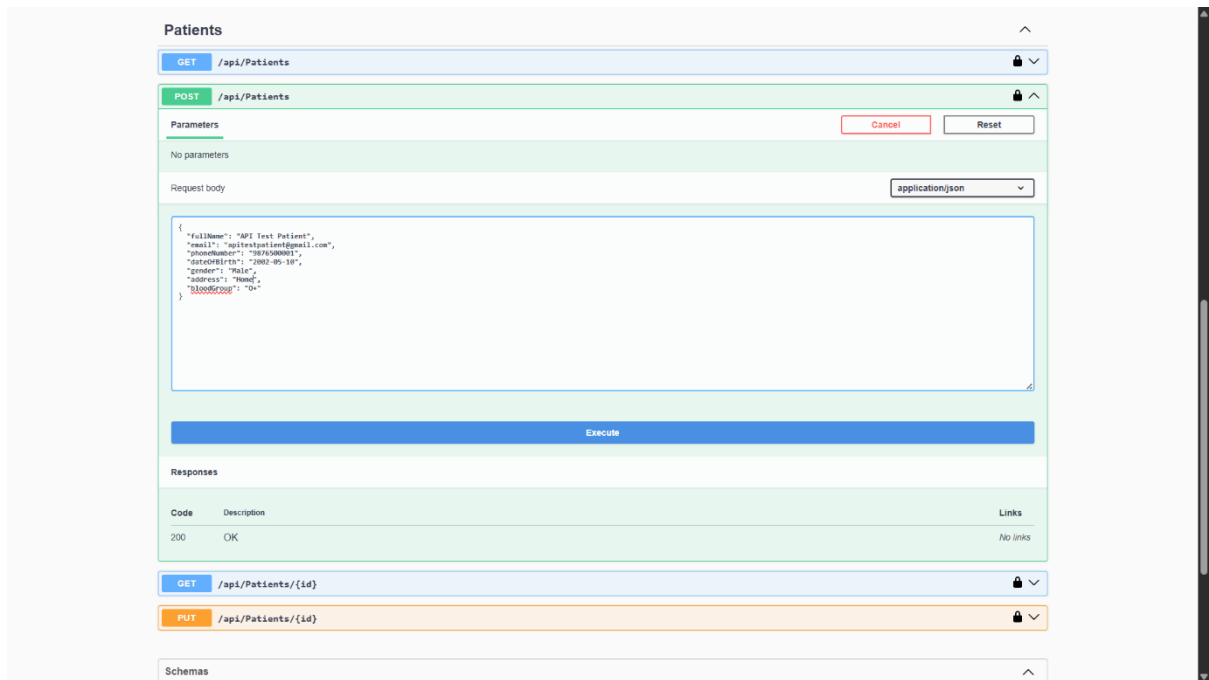


Figure: New patient created successfully through Web API.

GET /api/Appointments/{id}

PATCH /api/Appointments/{id}/status/{status}

Auth

POST /api/Auth/login

Doctors

GET /api/Doctors

GET /api/Doctors/{id}

Parameters

Cancel

Name Description

id * required
Integer (\$int32)
(path) 1

Execute

Responses

Code	Description	Links
200	OK	No links

Patients

GET /api/Patients

POST /api/Patients

GET /api/Patients/{id}

PUT /api/Patients/{id}

GET /api/Doctors/{id}

Parameters

Cancel

Name Description

id * required
Integer (\$int32)
(path) 1

Execute Clear

Responses

Curl

```
curl -X 'GET' -\
  "https://localhost:58295/api/Doctors/1" -\
  -H 'accept: */*' -\
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiI1MDkxOTY3IiwiaWF0IjoxNjU0MjU0MjU0LCJpc29uIjoiYm9keiJ9.eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiI1MDkxOTY3IiwiaWF0IjoxNjU0MjU0MjU0LCJpc29uIjoiYm9keiJ9
```

Request URL

https://localhost:58295/api/Doctors/1

Server response

Code Details

200

Response body

```
{
  "id": 1,
  "fullName": "Dr. Ananya Rao",
  "email": "ananya.rao@healthcare.local",
  "phoneNumber": "9876543210",
  "specialization": "General Physician",
  "specialtyId": 1,
  "isActive": true
}
```

Response headers

```
content-security-policy: default-src 'self'; frame-ancestors 'none';
content-type: application/json; charset=utf-8
date: Fri, 05 Jun 2026 20:01:15 GMT
referrer-policy: no-referrer
server: Restify
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	OK	No links

Figure: Doctor details retrieved by Doctor ID.

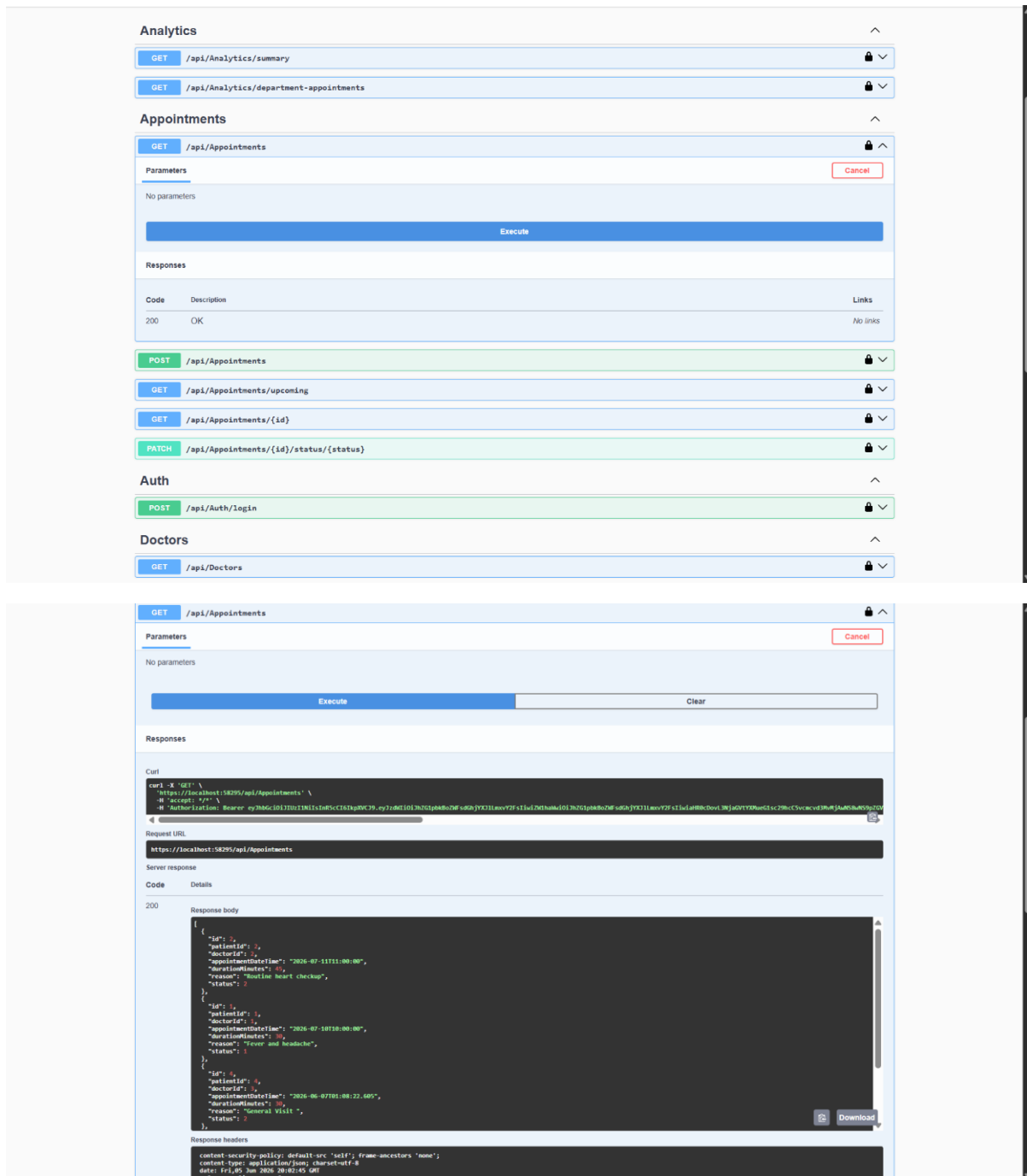


Figure: Appointments API returning appointment records from SQL Server.

The image displays a REST client interface with two panels. The top panel shows the configuration for a POST request to `/api/Appointments`. The request body is a JSON object: `{ "patientId": 1, "doctorId": 1, "appointmentDateTime": "2026-06-10T11:00:00", "durationMinutes": 30, "reason": "General health checkup" }`. The bottom panel shows the response, which is a 201 status code with a JSON body: `{ "id": 1, "patientId": 1, "doctorId": 1, "appointmentDateTime": "2026-06-10T11:00:00", "durationMinutes": 30, "reason": "General health checkup", "status": 1 }`. The response headers include `content-security-policy`, `content-type`, `date`, `location`, `referrer-policy`, `server`, `x-content-type-options`, and `x-frame-options`.

Top Panel: Request Configuration

- Method: **POST**
- URL: `/api/Appointments`
- Parameters: No parameters
- Request body: `application/json`
- Request body content:

```
{
  "patientId": 1,
  "doctorId": 1,
  "appointmentDateTime": "2026-06-10T11:00:00",
  "durationMinutes": 30,
  "reason": "General health checkup"
}
```
- Execute button

Bottom Panel: Response Details

- Response status: **201**
- Response body:

```
{
  "id": 1,
  "patientId": 1,
  "doctorId": 1,
  "appointmentDateTime": "2026-06-10T11:00:00",
  "durationMinutes": 30,
  "reason": "General health checkup",
  "status": 1
}
```
- Response headers:

```
content-security-policy: default-src 'self'; frame-src: none;
content-type: application/json; charset=utf-8
date: Fri, 05 Jun 2026 20:06:02 GMT
location: https://localhost:50295/api/Appointments/5
referrer-policy: no-referrer
server: Kestrel
x-content-type-options: nosniff
x-frame-options: DENY
```
- Execute and Clear buttons

Figure: Appointment created successfully using REST API.

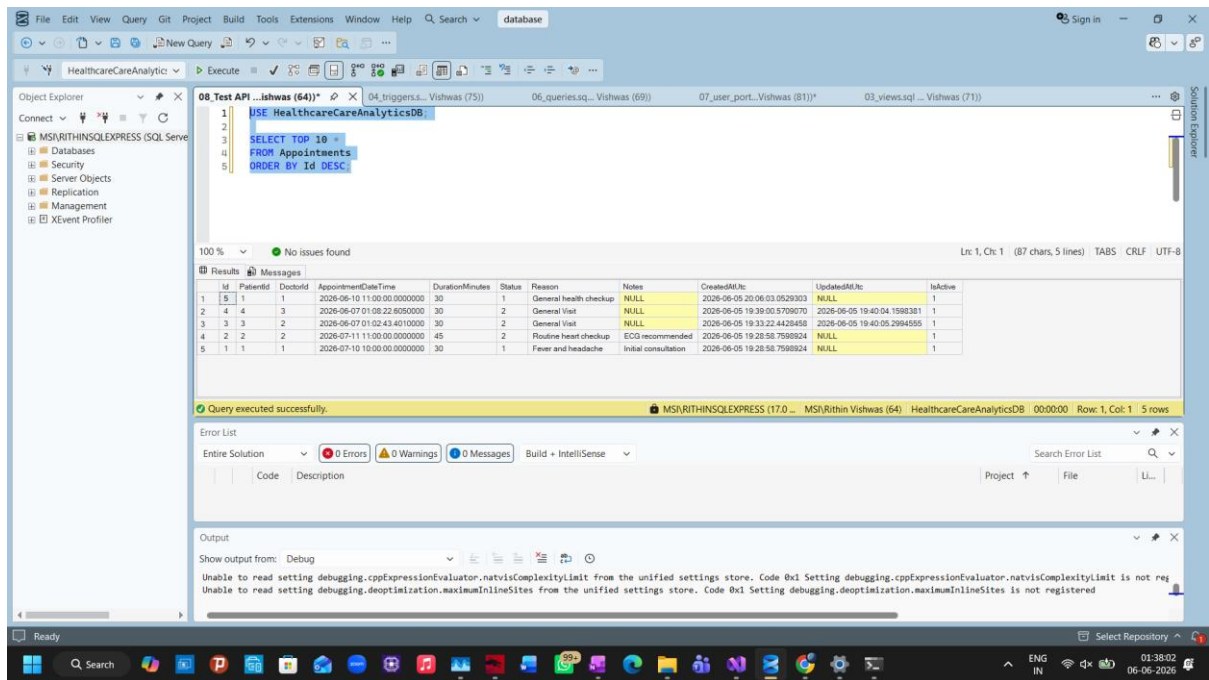


Figure: Appointment record verified in MS SQL Server after API execution.

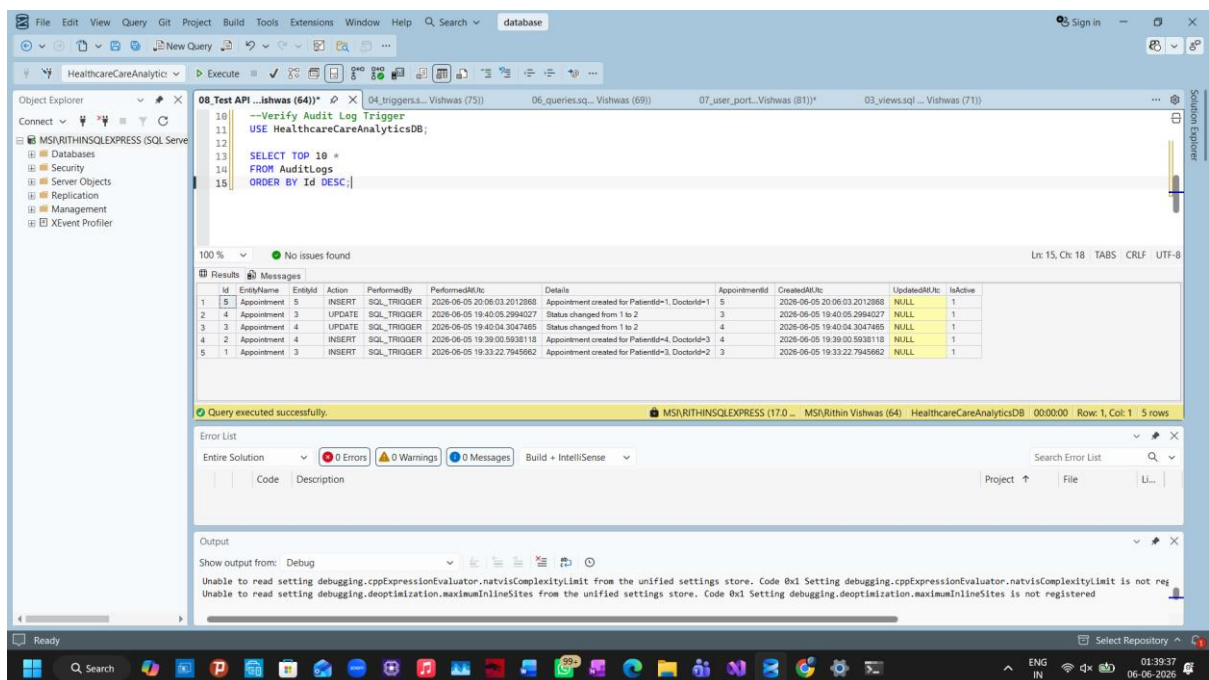


Figure: SQL trigger created an audit log after appointment insertion.

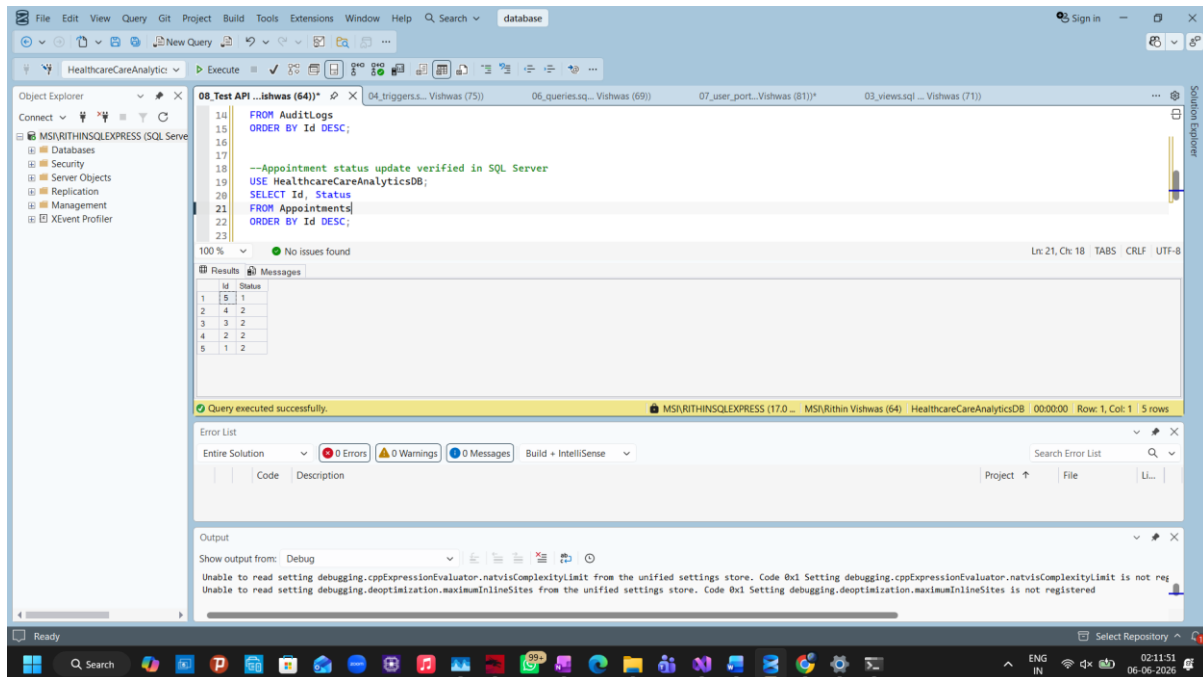


Figure: Appointment status update verified in SQL Server.

API Testing Explanation

Tested the Web API using Swagger. First, I started the Healthcare.API project and opened Swagger UI. Then I tested the login endpoint with admin credentials. After successful login, the API generated a JWT token. I used the token in Swagger Authorize using the Bearer format. Then I tested protected endpoints such as Patients, Doctors, and Appointments. I created an appointment through the API and verified the inserted data directly in SQL Server. I also checked the AuditLogs table to confirm that the SQL trigger executed successfully. Finally, I removed the token and tested a protected endpoint again, which returned 401 Unauthorized, proving that JWT security is working.

API demo flow:

Run API → Open Swagger → Login → Get JWT → Authorize → Test Patients → Test Doctors → Create Appointment → Verify SQL → Verify Audit Log → Test Unauthorized Access